

# Design Document — FindLostPuppy 🎨

| Document Metadata | Value | | :--- | :--- | | **Title** | System, UI/UX & Interaction Design Specification | | **Version** | 1.0.0 | | **Theme / Aesthetic** | Obsidian Titanium (Sleek Dark Theme with Vibrant Signal Accents) | | **Design Principles** | Zero Clutter, High Contrast Emergency Signals, Mobile-First Thumb Ergonomics | | **Target Viewports** | 320px (Compact Mobile), 390px (Standard Mobile), 768px (Tablet), 1024px+ (Desktop) |

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## 1. Design Philosophy & Aesthetic Pillars

FindLostPuppy is engineered for high-stress emergency scenarios where clarity, speed, and focus save lives.

**1. Obsidian Titanium Base:** Deep, non-reflective dark surfaces ( `#09090B` base background, `#18181B` surface cards, `#27272A` borders) provide high visual depth, minimize battery consumption on OLED displays, and eliminate eye strain in daytime or nighttime field searches. **2. Intentional Signal Colors:** - **Signal Emergency Orange** ( `#FF6600` / `#FF7711` ): Reserved for primary calls-to-action, active missing alerts, and crucial navigation anchors. - **Rescue Emerald** ( `#10B981` / `#059669` ): Reserved strictly for safe statuses, verified pet presence indicators, and completed checks. - **Alert Crimson** ( `#EF4444` / `#DC2626` ): Reserved for emergency missing declarations and critical destructive confirmations. **3. Ergonomic Thumb-Zone Navigation:** All primary actions (Quick Capture FAB, Next, Back, Modal confirmation buttons) are pinned or positioned within standard one-handed thumb reach. **4. Fluid Motion Physics:** Transitions use cubic-bezier easing ( `cubic-bezier(0.16, 1, 0.3, 1)` ) with micro-durations ( `180\text{ms}` - `350\text{ms}` ) to convey responsiveness without artificial lag.

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## 2. Design System Tokens & Tokens Matrix

```
`` | OBSIDIAN TITANIUM |
| Background: #09090B | Border
(Subtle): #27272A | | Card Glass: #18181B | Border (Glow): #FF6600/35 | | Text Prime: #F4F4F5 |
Text Secondary: #A1A1AA | | Accent
Warm: #FF6600 | Safe Green: #10B981 | | Accent Hover: #FF8533 | Danger Red: #EF4444 |
``
```

### 2.1 Color Palette Tokens

```
`css :root { / Surfaces / --bg-primary: #09090B; --bg-surface: #18181B; --bg-surface-elevated:
#27272A; --bg-card-glass: rgba(24, 24, 27, 0.88); --backdrop-blur: blur(16px);

/ Typography / --text-primary: #F4F4F5; --text-secondary: #A1A1AA; --text-tertiary: #71717A; --
text-inverse: #09090B;
```

```
/ Accents & Signals / --accent-orange: #FF6600; --accent-orange-hover: #FF8533; --accent-orange-glow: rgba(255, 102, 0, 0.28); --signal-green: #10B981; --signal-green-bg: rgba(16, 185, 129, 0.12); --signal-green-border: rgba(16, 185, 129, 0.35);
```

```
--signal-red: #EF4444; --signal-red-bg: rgba(239, 68, 68, 0.12); --signal-red-border: rgba(239, 68, 68, 0.35);
```

```
/ Boundaries & Radii / --border-subtle: rgba(255, 255, 255, 0.08); --border-focus: #FF6600; --radius-sm: 8px; --radius-md: 14px; --radius-lg: 20px; --radius-full: 9999px; }
```

## 2.2 Typography Hierarchy

- **Primary Font Family:** System UI stack with Outfit / Inter fallback:  
-apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, "Helvetica Neue", Arial, sans-serif
- **Scale:**
  - Display / Header 1 : clamp(1.5rem, 5vw, 2.25rem) (Weight: 800) - Header 2 : clamp(1.2rem, 3.5vw, 1.5rem) (Weight: 700) - Section Title : 1.125rem (Weight: 600) - Body / Inputs : 0.9375rem ( 15px , Weight: 400/500) - Caption / Meta : 0.8125rem ( 13px , Weight: 400)

## 3. UI Component Architecture & Key Controls

### 3.1 BackButton Component ( src/components/ui/back-button.tsx )

Designed to replace plain icon links with an interactive, tactile pill control:

- **Default State:** Dark pill container ( #27272A ) with visible "Back" text and a left-aligned arrow icon box.
- **Hover / Active State:** Arrow icon background animates from 25% to 100% width across 300ms , fading out the "Back" text and highlighting the sliding left arrow.
- **Usage Across Onboarding:** Positioned at the top-left of Location, Pet Choice, Pet Details, and Lost Dog Alert screens.

### 3.2 Animated AlignJustifyIcon ( src/components/ui/align-justify-icon.tsx )

A hamburger navigation trigger utilizing native SVG SMIL dashoffset animations:

- Composed of three sequential stroke animations:
  - Top bar draws in 0.2s - Middle bar starts at 0.2s and completes at 0.4s - Bottom dual lines start at 0.4s and complete at 0.6s
- Integrated into the mobile top header ( .mobile-menu-trigger-btn ). Smoothly transitions between the hamburger icon and X close icon when the mobile drawer toggles.

### 3.3 Dashboard Category Status Cards

Three high-impact cards with live status counts and badges: 1. **Sighted Missing Pets:**

Amber/Orange glow border ( --accent-orange-glow ), representative pet avatar or icon, count badge of recent neighborhood sightings. 2. **Pets at Home:** Emerald border ( --signal-green-

border ), safe check indicator, displays owner's registered pet name and health/safe badge.  
3. **Pets Missing:** Crimson border ( --signal-red-border ), pulsing alert ring, displays active missing alerts count.

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## 4. Screen-by-Screen User Flows & Layouts

1. Splash / Launch — Dark native splash → Animated FindLostPuppy logo (Web/Capacitor)  
2. Authentication (EmailAuthPage) — Clean Google Sign-In card + Guest preview option 3. Inauguration & Gratitude (First-Time User Only) — Sonu memorial dedication → "Begin Rescue Journey"  
4. Pet Parent Contact (OwnerOnboardingPage) — Full Name, Indian Phone (10 digits), Photo Upload, Read-only View on completion  
5. Location Onboarding (LocationOnboardingPage) — Resilient Auto-Detect Button + State → District → Mandal → Village Selectors  
6. Pet Choice & Details (OnboardingChoicePage & DogOnboardingPage) — "Pet Registered" / "You already had a pet" → Full Breed/Age/Color/Photo Profile  
7. Community Dashboard (DashboardPage) — 3 Category Cards → Single Modal Explorer → Grouped Sighting Drawer

### 4.1 Modal Visibility & Layering Invariants

To eliminate mobile clutter and context confusion:

- **Single Modal Invariant:** Only one modal dialog may be visible on screen at any time.
- Selecting a category card opens the **Category List Modal**.
- Selecting a pet or sighting from the list **hides** the category list modal and opens the **Detail Modal**.
- Closing the detail modal **restores** the category list modal at the exact previous scroll position.
- Sighting details group all sightings for a single missing dog under numbered tabs: Sighting 1 , Sighting 2 , etc.

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## 5. Responsive Design & Breakpoint Strategy

| Breakpoint    | Target Form Factor                                              | Layout Behavior                                                                                                     |
|---------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| < 360px       | Small devices (e.g., iPhone SE, compact Androids)               | Single-column cards, compact button paddings ( 10px 14px ), font scale clamp, zero horizontal scroll.               |
| 360px - 480px | Standard smartphones (e.g., iPhone 14/15, Samsung Galaxy, iQ00) | Full mobile layout, sticky bottom FAB for quick capture, touch target minimum $\geq 44 \times 44 \text{px}$ .       |
| 481px - 768px | Foldables & small tablets                                       | 2-column card grid where appropriate, centered container with max-width 460px for onboarding forms.                 |
| > 768px       | Tablets & Desktops                                              | Fixed collapsible sidebar navigation ( SidebarNav ), side-by-side dashboard panels, max-width 1200px app container. |

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## 6. Accessibility (a11y) & Usability Standards

1. **Color Contrast:** All text tokens meet WCAG 2.1 AA standards (minimum 4.5:1 ratio for body text against #18181B surfaces; minimum 3.0:1 for large headlines and interactive icons).  
2. **Keyboard Navigation:** Every button, selector, card, and modal backdrop supports explicit :focus-

visible outline rings with Enter and Space activation keys. 3. **Screen Readers:** All icon-only buttons include descriptive aria-label and title attributes (e.g., aria-label="Toggle navigation menu" , aria-label="Back to Location" ). 4. **Motion Safety: Respects** prefers-reduced-motion: reduce` by disabling non-essential SMIL/CSS transitions and celebratory confetti animations.